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Top Skills

Cursors
Claude Code Subagents
Back-End Web Development

Languages

English (Native or Bilingual)
Hindi (Limited Working)

Certifications

AZ-300: Microsoft Azure Architect Technologies
AMCAT Certified Software Engineer - IT Services
SQL (Basic) Certificate
SQL (Intermediate) Certificate
Python (Basic) Certificate

Honors-Awards

Amul vidya shree

Publications

OSI Model Demystified
Azure Cloud Migration (Step 2)
Kubernetes Architecture Demystified
What's new with Azure Backup
Azure structure explained

Nadeem Khan

Senior Software Engineer | Agentic Data Systems | Distributed Systems | Applied AI
Chicago, Illinois, United States

Summary

I'm a backend and distributed systems engineer focused on building and owning high-risk, high-scale backend platforms where correctness, determinism, and operational reliability are non-negotiable.

My work centres on distributed execution systems, query runtimes, and data platforms operating under real production constraints such as high throughput, strict latency budgets, partial failures, and sustained cost pressure. I approach AI as an engineering problem first. Clear lifecycles, explicit interfaces, controlled execution, and observability matter more than model novelty. I use generative AI where it creates measurable production or business value, not as a substitute for system design.

I've designed and operated production systems processing 10TB + of data per day and tens of millions of events per day, with a strong emphasis on fault tolerance, backpressure, idempotency, and recoverability. Across these platforms, I've delivered significant latency improvements, approximately 40% reduction in cloud costs, and over 99% production reliability, supported by deep, customer-facing observability using OpenTelemetry.

One representative system I've built and owned end-to-end is a production-grade natural language-to-SQL platform for safely querying real enterprise databases. Rather than relying on prompt-driven text generation, the system implements a multi-stage execution lifecycle including planning, validation, compilation, and execution, with schema-aware validation, deterministic SQL generation, sandboxed execution, and full end-to-end tracing. I owned the architecture, runtime design, performance tuning, and operational reliability of this system in production.

Beyond individual systems, I take responsibility for technical direction on core backend platforms. This includes leading design reviews, making trade-offs in ambiguous situations, setting reliability and execution standards, and mentoring engineers building production-grade distributed systems.

Experience

EvolutionIQ

Senior Software Engineer

March 2026 - Present (6 months)

New York, United States

- Designed and shipped a product extension that surfaces critical claim information directly to adjusters, eliminating manual parsing of demand packages and accelerating claims workflows.
- Led a cross-stack refactor spanning frontend and backend codebases, improving readability and maintainability to cut technical debt and speed up new-engineer onboarding.
- Introduced SQL query tagging to deliver end-to-end performance visibility across jobs and routes, pinpointing execution-time hotspots and driving targeted optimizations.

Crowe

4 years 8 months

Senior Software Engineer

July 2023 - Present (3 years 2 months)

Chicago, Illinois, United States

Senior Software Engineer and Systems Architect building large-scale, cloud-native execution platforms across distributed systems and data architecture. I focus on production-grade, observable systems with deterministic execution and operational reliability.

I design multi-stage SQL execution pipelines and agent-assisted planning systems using LangGraph on Azure, with emphasis on traceability and correctness. I also own and operate a Distributed SQL Execution Platform, a production system for safely querying enterprise databases at scale.

Key Contributions

- Built a deterministic SQL execution platform with planning, AST validation, sandboxed execution, and end-to-end tracing, supporting Postgres, MySQL, SQL Server, and SQLite.
- Designed DAG-based execution runtimes with durable orchestration, fine-grained parallelism, connection-pool-aware backpressure, and fault-tolerant recovery using persistent state and idempotency. Powered CDC pipelines processing 60M+ PostgreSQL change events/day and batch workloads of 10TB+/day.
- Led rule-based SQL rewrites (join reordering, predicate pushdown, query decomposition) and execution-aware storage layouts, reducing pipeline runtime from 4–9 hours to 2–2.5 hours by minimising intermediate cardinality.
- Implemented deterministic migration workflows with schema validation and watermark-based incremental ingestion for consistent, replayable enterprise data migrations.
- Built OpenTelemetry-based observability with per-stage latency dashboards and execution traceability, sustaining 99%+ reliability while reducing ETL runtime from 8 hours to 2.5 hours and cutting cloud costs by ~40%.

Tech: Python, Java, SQL, LangGraph, Azure, Docker, Kubernetes, Kafka, PostgreSQL, Azure Synapse, OpenTelemetry

Cloud Senior Engineer

January 2022 - June 2023 (1 year 6 months)

Chicago, Illinois, United States

Owned the architecture, implementation, and 24×7 reliability of a real-time change data capture (CDC) platform synchronizing PostgreSQL operational data with Salesforce at scale.

Key Impact

- Scaled ingestion to 60M+ row changes/day by combining PostgreSQL logical replication (wal2json) with Debezium on Kafka Connect, distributing events across multiple partitions to maximize parallelism, throughput, and fault tolerance.
- Reduced Salesforce Bulk API usage by 80% and improved end-to-end latency by 40% by batching CDC events and orchestrating high-throughput, idempotent upserts via Azure Logic Apps using External ID-based writes.
- Designed and deployed a containerized Spring Boot consumer service with concurrent processing, dead-letter queues, idempotent upsert logic,

and exponential backoff retries, ensuring exactly-once-like behavior at the application layer.

- Established zero-touch operations by exporting Kafka JMX metrics to Azure Application Insights, implementing lag-based alerting, auto-scaling AKS consumer pods, and delivering blue-green deployments through Azure DevOps CI/CD pipelines.

Technology Stack

PostgreSQL (wal2json), Debezium, Apache Kafka, Kafka Connect, Java, Spring Boot, Docker, Azure Kubernetes Service (AKS), Azure Logic Apps, Salesforce Bulk API, Azure Application Insights, Azure DevOps

Boston University

10 months

Racial Data & Policy Student Manager at Centre of Antiracist Research
May 2021 - February 2022 (10 months)

Boston, Massachusetts, United States

As the overseer of product development and lifecycle, I led the creation of a website aimed at providing people with a comprehensive understanding of the current situation of different races in the US. The primary objective was to shed light on important societal aspects such as excess mortality rate, unemployment rate, homeownership rate, and various other covariates specific to different racial and ethnic groups.

To achieve this, I developed interactive visualizations using ReactJS and D3. These visualizations effectively presented data in a clear and engaging manner, allowing users to explore and compare the disparities across different races and ethnicities. Through these visuals, users were able to gain insights into the social, economic, and health conditions of various communities within the United States.

By carefully curating relevant data and leveraging the capabilities of ReactJS and D3, we created a powerful tool for raising awareness and promoting informed discussions about racial disparities. The website's product development and lifecycle management were focused on ensuring accurate data representation, maintaining data integrity, and providing an intuitive user experience.

Throughout the process, I collaborated closely with stakeholders, data analysts, and designers to ensure alignment with the project's goals and

user requirements. This involved iterative development, user testing, and incorporating feedback to continuously enhance the website's functionality and usability.

In summary, I oversaw the end-to-end development and management of a website dedicated to providing insights into the current situation of different races in the US. By utilizing ReactJS and D3, I developed interactive visualizations that effectively conveyed data related to various socio-economic indicators, contributing to a better understanding of racial disparities.

Teaching Assistant

May 2021 - January 2022 (9 months)

Boston, Massachusetts, United States

In the course MET CS 677: Data Science with Python, my responsibilities include conducting group office hours to assist students, clarifying conceptual questions through one-on-one sessions, providing individual feedback on assignments, and offering recommendations for best programming logic practices.

During group office hours, I create a supportive environment where students can ask questions, seek clarification, and engage in discussions related to the course material. This allows me to address common challenges, reinforce key concepts, and provide additional guidance as needed.

In one-on-one sessions, I work closely with students to clarify any specific conceptual questions they may have. By offering personalized explanations and examples, I ensure that students have a clear understanding of the topics covered in the course.

When evaluating assignments, I provide constructive feedback to help students improve their work. This includes identifying areas of strength and areas that require further development. Additionally, I recommend best programming logic practices to enhance the quality and efficiency of their code.

As a grader for a graduate-level course, I am responsible for evaluating and assigning grades to student assignments. I adhere to the course guidelines and grading criteria to ensure fairness and consistency in the evaluation process.

Overall, my role in MET CS 677: Data Science with Python involves supporting students through group office hours, one-on-one sessions, feedback on assignments, and grading. I strive to provide a positive and enriching learning experience for students as they navigate the course content and develop their skills in data science using Python.

Crowe

Senior Cloud Software Engineer

March 2018 - December 2020 (2 years 10 months)

India

As a Backend Team Lead, I supervised a team of 5 members, ensuring their progress and providing constructive feedback. My primary responsibility was to manage and orchestrate the product development of a SaaS platform, focusing on improving visibility in outside processing.

To achieve enhanced scalability, I designed and implemented a horizontally scalable Microservice architecture using Spring, Docker, and Kubernetes. By utilizing the Horizontal Pod Autoscaler, I set the target average CPU utilization of all pods at 70%. This approach allowed us to efficiently allocate computing resources and accommodate increasing workload demands.

Through effective architectural planning and programming, we were able to achieve a scalable and robust infrastructure that seamlessly handled high volumes of data and user requests. The use of Spring, Docker, and Kubernetes enabled easy deployment, management, and scaling of Microservices, ensuring optimal performance and reliability.

In addition to technical responsibilities, I also facilitated communication and collaboration within the team, promoting agile methodologies and ensuring alignment with project objectives. By overseeing the product development process, I ensured timely delivery and adherence to quality standards.

Overall, as a Backend Team Lead, my role involved managing a team, orchestrating product development, and implementing scalable infrastructure solutions to optimize the performance and visibility of the SaaS platform.

Sports Social

Full Stack Developer

June 2017 - March 2018 (10 months)

India

I developed the initial version of a blog website for the organization using Angular, NodeJS, and MySQL. This involved designing and implementing the frontend using Angular, creating the backend using NodeJS, and utilizing MySQL for the database management.

Additionally, I worked on enhancing specific sections of the organization's main website. This involved implementing new features, improving existing functionalities, and ensuring a seamless user experience.

By leveraging Angular, NodeJS, and MySQL, I was able to deliver a robust and efficient blog website, enabling the organization to share content and engage with their audience effectively.

Education

Boston University

Master of Science - MS, Computer Science · (January 2021 - May 2022)

Krishna Institute of Engineering & Technology

Bachelor of Technology (BTech), Electronics and Communications Engineering · (2013 - 2017)